INTRODUCTION TO HYPOTHESIS TESTING

Learning Objectives

- Recognise the steps required for a Hypothesis test from the set up, assumptions, testing, and interpretation of p-values
- Identify a variety of tests and when they are used
- Evaluate the output of tests from an appropriate software tool

STATISTICAL INFERENCE: STEPS TO TAKE WHEN TESTING HYPOTHESES

Summary of Steps we will take:

Define any notation or abbreviations

State the null and alternative hypotheses – how you think the **population** behaves: H_0 and H_1 with the value(s) of any parameters and a descriptive sentence to help you interpret the results later. Identify if the test is one or two tailed.

Decide the **significance level**, α. (P-Value) 0.05

State that you are **assuming the null hypothesis is true** and state any logical deductions from this (e.g. the assumed distribution of the population).

Use a sample to calculate a **test statistic or p-value** based on the assumption that the null hypothesis is true.

Check **if** the test statistic is extreme or if the p-value is low – if either is the case you can **reject the assumption** that H_0 is true; the result from the sample is **significant** and conclude (in context) that there **might** be enough evidence to suggest the alternative is true.

SIGNIFICANCE LEVEL

The **significance level**, α.

How unlikely does something have to be before we start to wonder if we have been making bad assumptions?

We can't set it at 0% or we would *always* stick with our initial assumption and there would be no point doing the test!

But it also represents the chance of switching to an alternative hypothesis incorrectly – it's the chance of a **false positive**.

The level should be decided **before** calculations on the sample – to avoid scientific bias.

T-TEST: AN EXAMPLE OF A/B TESTING

Basic idea: Are the means of two samples similar or different? If they are similar they could come from the same population.

Examples:

- Is the average sentiment value from before and after a marketing campaign the same or have we made a positive impression? (1 Tailed)

Note: This is an example of **1 tailed**: specifically have we made a positive impact rather than just, is it different which would be **2 tailed**.

- Is the average investment performance different if we change strategy?(2Tailed – Not specific about up or down)

A/B testing can be used to compare before vs after, or distinct groups in the same time frame.

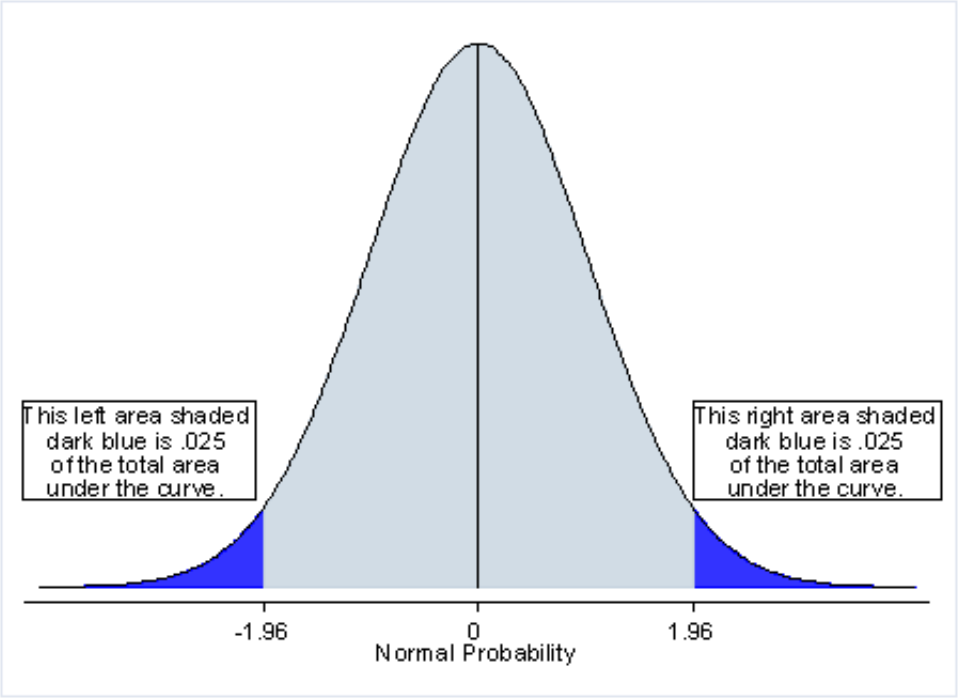
ONE AND TWO TAILED T-TESTS

Two Tailed Test (Non-Directional) looks for “is the result **different”** for example below(1 tail) a value or above(2 tail) a value but is not concerned with which.

One Tailed Test(Directional) looks specifically for the result to be, for example **below** or **above ,**basically it looks to prove just one of these results

T-Tests statistically show if there is a significant difference between the means of two sets of values.

TWO TAILED T-TEST



The tails of the t-distribution are the areas where the test statistic is most unlikely to fall if the null hypothesis is true. If the calculated t-statistic falls within the tail regions, it suggests that the observed data is too extreme to be explained by random chance alone, and the null hypothesis is rejected.

T-TEST EXAMPLE SET UP

Group A: all the investments follow the standard strategy

Group B: all the investments use the new strategy

H_0: The population means of group A and B are the same – i.e. it makes no difference which strategy is used.

H_1: The population means of group A and B are different – i.e. it does make a difference which strategy is used.

This is 2 tailed.

Significance level: 5% = 0.05. As this is 2 tailed, the significance level is split when comparing to the p-value later: 0.025

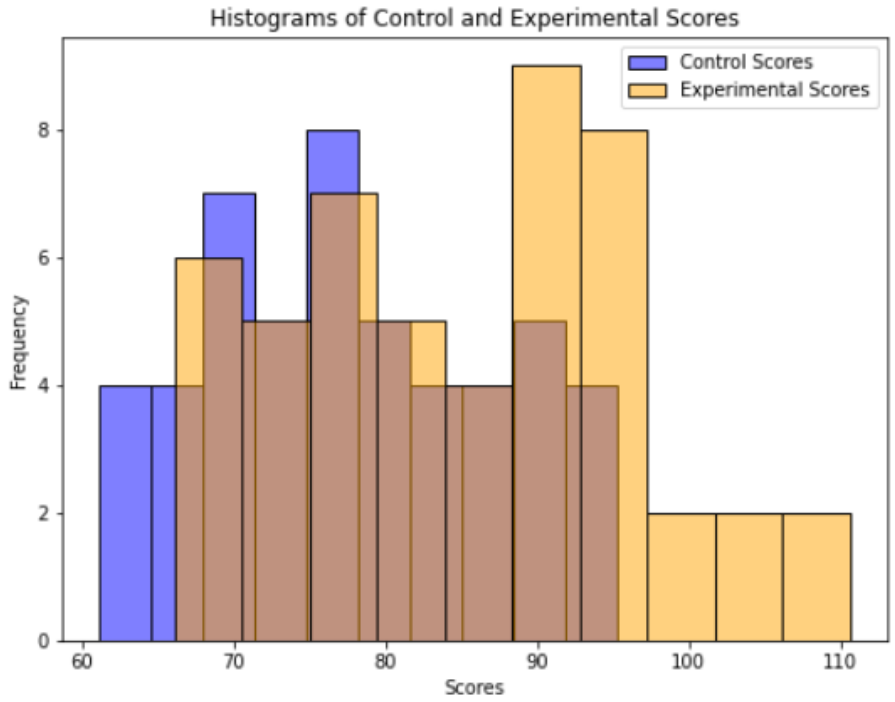
Assume the null hypothesis is true – this is done automatically by the tool or programming language you will use.

T-TEST VISUALISED AND IN CODE

Are the these different enough or could they have come from the same population?

p-value: What's the chance the samples behave this way if the assumption is correct (that they come from the same population).

We can use tools like the Data Analysis Toolpak in Excel, Python with a package like scipy.stats, or R to calculate the p-value. Underlying code often uses the normal distribution to calculate probabilities and work out overlapping areas.



CONDUCTING A T-TEST PYTHON

- Import pandas as pd and from scipy import stats.
- Use pd.read_csv() to save the data in investment_performance.csv as df, a pandas dataframe.
- Use stats.ttest_ind() with each of the two columns from the dataframe addressed in the brackets.
- Compare the p-value to half the significance level (since it’s a two tailed test)
- Conclude: What to do about the assumption that was made? (That the null hypothesis is true).
- Conclude in context: What does that mean we have evidence for in this particular context?

SOLUTION

PYTHON

```python
import pandas as pd
from scipy import stats

csv_filename = 'investment_performance.csv'
df = pd.read_csv(csv_filename)
stats.ttest_ind(df['Control Scores'], df['Experimental Scores'])
```

```
Ttest_indResult(statistic=-3.496431755693145, pvalue=0.0007101500664096571)
```

Comparison: 0.000710 (3 s.f.) < 0.025

This is very unlikely to have happened if the assumption was correct.

The result is significant. Reject the null hypothesis and therefore accept the alternative.

In Context:

There seems to be enough evidence to suggest there is a difference between the two investment strategies.

Notice that the conclusion does not promise certainties!

OTHER COMMON TESTS

- Types of T-test and A/B testing:
- Correlation: comparing two numerical columns to look for links. Assume there is no correlation in the population, is the sample strongly correlated enough to suggest otherwise?
- Chi-Squared: comparing categorical columns to look for links. Assume there is no association in the population, are the observed vs expected frequencies different enough to suggest otherwise?
- ANOVA: comparing the means of more than 2 groups. Assume all groups come from the same population, are any of the means different enough to the others to suggest otherwise (downside: won't know which group or groups stand out)?

LEARNING CHECK

Think about your answers to these questions

- What are the steps you need to take when doing statistical tests or experiments?
- What could you use A/B testing for in your organisation or sector?
- LO3 question

HOW DID YOU GET ON?

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